

Branching Logic

Project: Transparent Book Recommender

1. Your mission

Your learning target

I can trace the selected branch in an if elif else chain.

I can use indentation to identify which statements belong to a branch.

I can test values at and around a decision boundary.

Remember before reading

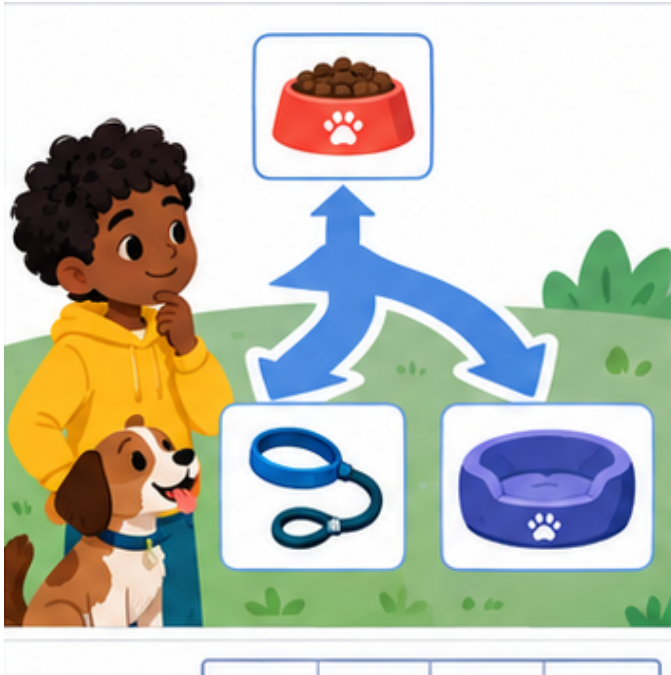
Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Decompose a Goal. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the evidence shows was wrong.

Week 3: Branching Logic

Ordered conditions determine which path a system takes.



THINKER MISSION

Build and trace a three-branch pet-care algorithm.

Thinker name and date:

Write or draw your reasoning and evidence.

Readiness for the connected explanation

Does minutes ≤ 12 accept exactly 12? How does $<$ differ?

This week's end goal

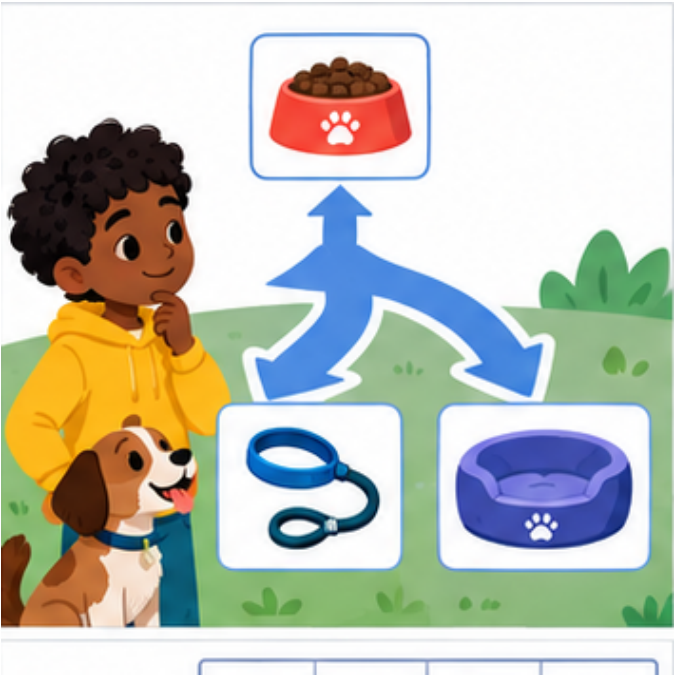
By the end, I can order branches so each input receives exactly its intended category.

2. Notice the evidence

PICTURE STORY

Notice $\rightarrow$ Predict $\rightarrow$ Explain

Begin with visible evidence. Do not explain more than the picture and information support.



MYSTERY

The learners must use branching logic to solve a problem. What information do they have? What decision or calculation do they make? What would count as evidence that it worked?

Three observations (what you can point to):

Write or draw your reasoning and evidence.

My prediction and the clue supporting it:

Write or draw your reasoning and evidence.

A closer explanation

A fictional activity sorter labels a supplied integer duration. Negative values are Invalid; values from zero through 12 are Short; larger values are Long. The labels describe duration only. Write this requirement before the program so that a test has an expected answer independent of the code.

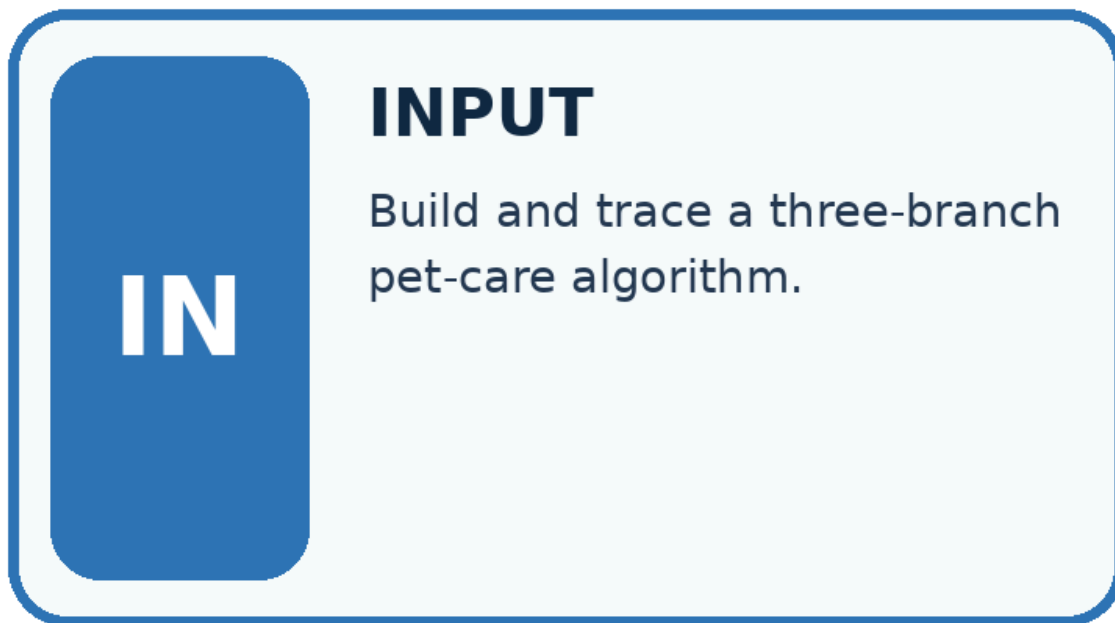
An if statement selects a block when its condition is true. An elif is considered only if earlier conditions in the same chain were false. An else handles the remaining cases. Colons end the condition lines. Use four spaces for the indented statements and keep related branches aligned.

3. See how it works

VISUAL MECHANISM

Input → Process → Output

Trace the information. At each stage, ask what changed and how you could test it.





PROCESS

Ordered conditions determine which path a system takes.





OUTPUT

Build and trace a three-branch pet-care algorithm.

Which stage could fail even when the other two are correct? Explain:

Write or draw your reasoning and evidence.

A closer explanation

One chain chooses one label

Input · Checks in order · Selected label

Input: -1

Checks in order: Below zero is True

Selected label: Invalid

Input: 12

Checks in order: Below zero False; <= 12 True

Selected label: Short

Input: 13

Checks in order: Both checks False

Selected label: Long

Later branches in this chain are skipped after a condition succeeds.

```
minutes = 12
if minutes < 0:
    label = "Invalid"
elif minutes <= 12:
    label = "Short"
else:
    label = "Long"
print(label)
```

The output is Short. For 12, the first condition is false and the second is true. The else block is skipped. `print(label)` is not indented, so it runs after the selected branch. Each path assigns label before this line reads it. Change only minutes and save before each fresh run.

Connect the picture and the explanation

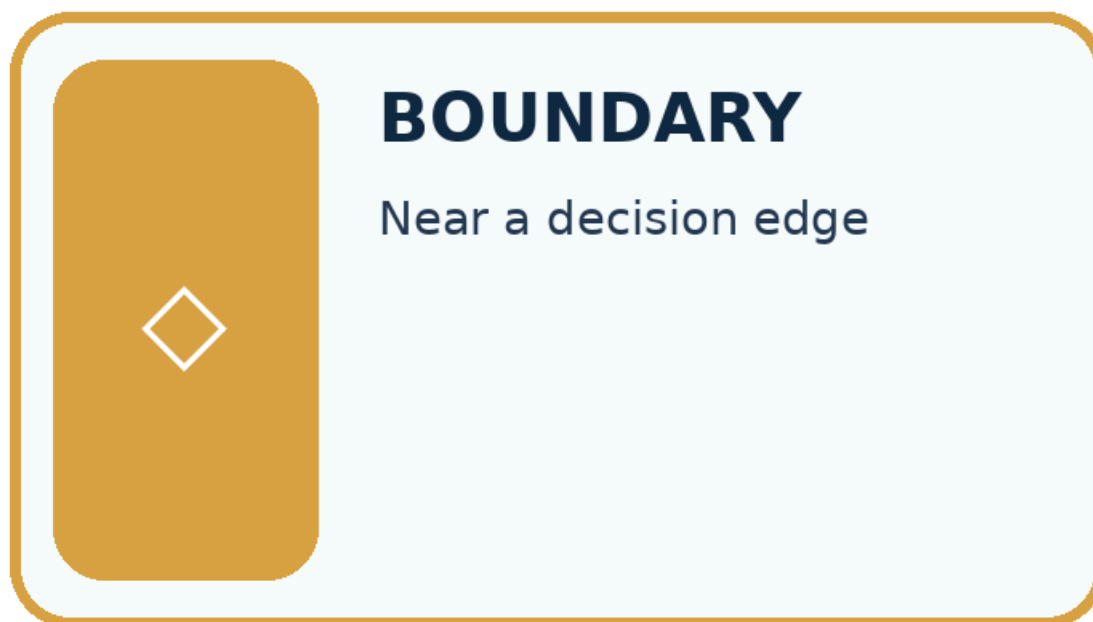
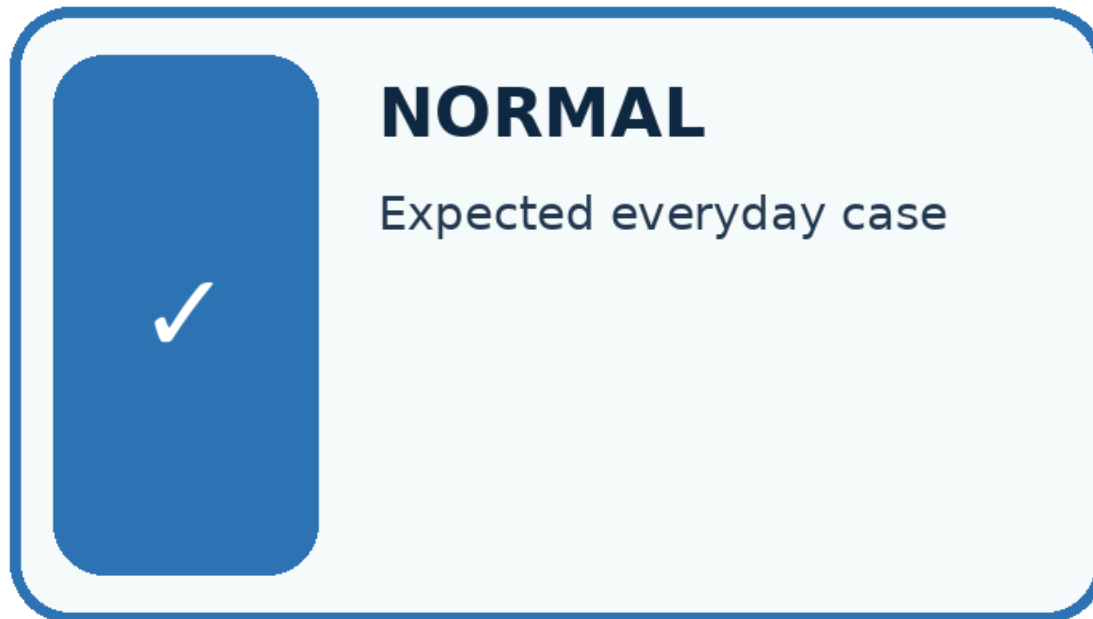
Point to the input, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

4. Compare the cases

CASE GALLERY

Four Tests, Four Kinds of Evidence

A system is not understood until it survives more than one comfortable example.



NORMAL CASE: predict and name evidence. BOUNDARY CASE: predict and name evidence.



MISSING

Needed information absent



TRANSFER

Fresh situation or user

MISSING INFORMATION: predict and name evidence. TRANSFER CASE: predict and name evidence.

Which case is most likely to reveal a hidden assumption? Why?

Write or draw your reasoning and evidence.

Change one thing in the pictured case

Change: Move `minutes <= 10` to the first branch.

Prediction: -1 and 0 wrongly receive Short.

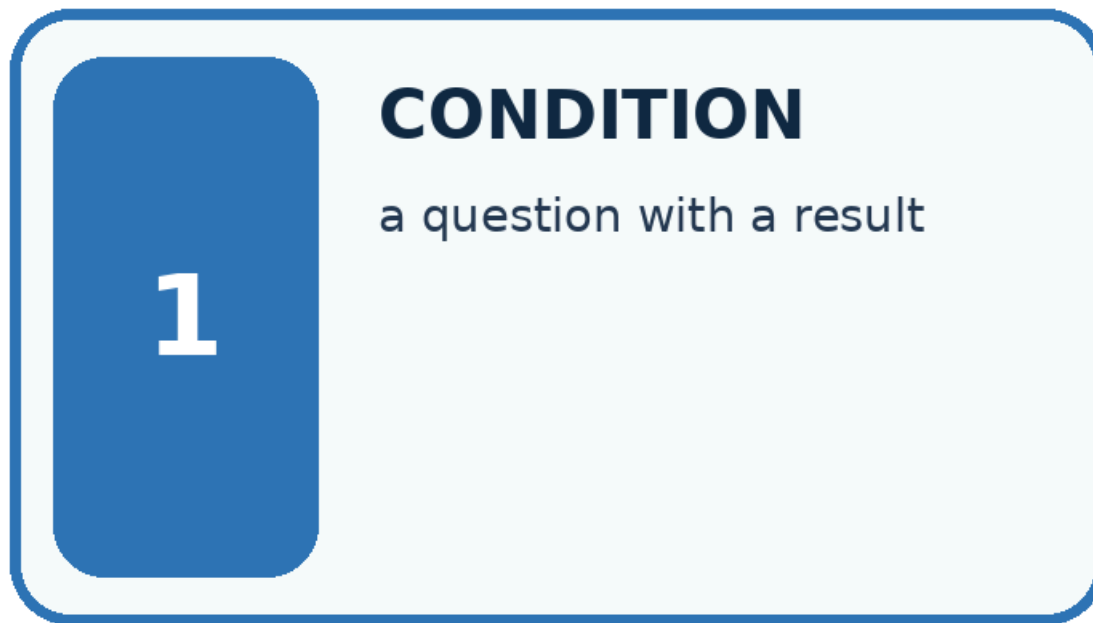
Reason: Both satisfy the earlier comparison before their intended checks are reached.

5. Words that unlock the idea

VOCABULARY LAB

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.



2

BRANCH

one path chosen after a condition

3

IF/ELSE

a two-way decision structure

4

ORDER

which condition is checked first

Word · My own example · Not this / boundary

Word: condition

My own example: A statement evaluated as true or false to help select an action.

Not this / boundary: _____

Word: branch

My own example: One path taken when a condition selects among alternatives.

Not this / boundary: _____

Word: if/else

My own example: A programming structure that chooses a path according to whether a condition is true or false.

Not this / boundary: _____

Word: order

My own example: The sequence in which steps are performed or items are read.

Not this / boundary: _____

Words used in the closer explanation

Word · Meaning

Word: Indentation

Meaning: Leading spaces that identify a Python code block

Word: Boundary

Meaning: A value where a classification may change

Word: Regression check

Meaning: A previous test rerun after an edit to detect a new failure

6. Follow a worked example

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete evidence.

WEEK 3 • WORKED EVIDENCE

- 1 IF pet is hungry → feed.
- 2 ELSE IF pet needs exercise → walk or play.
- 3 ELSE → provide a quiet rest space.
- 4 Boundary test: what if hungry and needs exercise? The first true branch wins.

1. IF pet is hungry → feed.
2. ELSE IF pet needs exercise → walk or play.
3. ELSE → provide a quiet rest space.

4. Boundary test: what if hungry and needs exercise? The first true branch wins.

Recalculate, retrace, or restate the reasoning in your own representation:

Write or draw your reasoning and evidence.

CHECK Did the evidence answer the exact question? Did the total, path, or source record stay consistent?

A closer explanation

Input minutes · Expected label · Reason for this test

Input minutes: -1

Expected label: Invalid

Reason for this test: Negative input rejection

Input minutes: 0

Expected label: Short

Reason for this test: Lowest valid duration

Input minutes: 11

Expected label: Short

Reason for this test: Just below the upper boundary

Input minutes: 12

Expected label: Short

Reason for this test: Exact upper boundary

Input minutes: 13

Expected label: Long

Reason for this test: Just above the boundary

These tests exercise every branch and the equality case. A change from ≤ 12 to < 12 is exposed by 12; testing 11 alone misses it. Branch coverage means every branch was exercised, but it does not prove every condition is correct.

Suppose the program uses < 12 instead of ≤ 12 . What is printed for 12, why does that violate the written rule, and which earlier test should still pass after repair?

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Branching Logic

A fictional reading-time labeler uses one Python if/elif/else chain.

The facts we will use

Whole minutes below 0: Invalid; 0: Empty; 1–10: Short; above 10: Long.

Order checks as $\text{minutes} < 0$, $\text{minutes} == 0$, $\text{minutes} \leq 10$, otherwise. Print the assigned label once.

Branching Logic

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Negative check

-1 takes Invalid and skips later branches.



2. Zero check

0 takes Empty, not Short.



3. Inclusive short range

1 and 10 take Short.



4. Remaining inputs

11 takes Long; only the selected branch assigns the label.

CHANGE ONE THING

Move minutes ≤ 10 to the first branch.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Negative check

What happens and why: -1 takes Invalid and skips later branches.

Step: 2. Zero check

What happens and why: 0 takes Empty, not Short.

Step: 3. Inclusive short range

What happens and why: 1 and 10 take Short.

Step: 4. Remaining inputs

What happens and why: 11 takes Long; only the selected branch assigns the label.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

The third comparison alone would include negative values and zero, but earlier branches already handle them. An elif chain selects its first true branch. Multiple independent if statements would have different behavior and could overwrite an earlier label. Boundary traces show why order belongs to the specification.

What this example does not establish

The example assumes whole numeric minutes; strings, missing values and fractional values need a separate declared input contract.

Read the labelled picture: Give expected labels for -1,0,10,11 and identify the equality boundary for Short.

Change one thing: Move $\text{minutes} \leq 10$ to the first branch. Predict the result before checking the explanation. What remains the same?

7. Find and repair a mistake

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak evidence.



COMPARE • DIAGNOSE

Weak idea: All true conditions run even in an if/else-if chain. Strong move: locate evidence, change one cause, and retest.

Evidence record · Expected · Actual · Likely cause / next test

Evidence record: normal case

Expected: works

Actual: _____

Likely cause / next test: _____

Evidence record: boundary case

Expected: stated rule

Actual: _____

Likely cause / next test: _____

Evidence record: missing input

Expected: safe response

Actual: _____

Likely cause / next test: _____

Evidence record: fresh transfer

Expected: generalizes

Actual: _____

Likely cause / next test: _____

Repair one cause. Why is this repair better than changing several things?

Write or draw your reasoning and evidence.

A closer explanation

For each run, record input, expected label, conditions actually checked and observed label. Mark skipped conditions as skipped rather than pretending the program evaluated them. After a repair, rerun earlier cases as well as the failing one. Week 6 will repeat a block and track changes across iterations.

A learner tests -1, 2 and 20 and concludes the sorter is completely correct because all three branches ran. Explain what important case is missing and why branch coverage alone is insufficient.

8. Practise together

LOGIC STUDIO

Trace Before You Run

Block code, pseudocode, tables, and diagrams are different ways to expose the same reasoning.



TRACE THE LOGIC

Predict the result. Trace every condition or calculation. Compare expected with actual. Explain the first difference.

SET evidence_record PREDICT expected_result RUN one normal case and one boundary case
COMPARE actual_result with expected_result IF different: locate first cause; change one factor; RETEST
EXPLAIN result, limitation, and next safe step

Trace one input. Which line changes the state or chooses a path?

Write or draw your reasoning and evidence.

A closer explanation

```
minutes = -1

if minutes < 0:
    label = "Invalid"

if minutes <= 12:
    label = "Short"

else:
    label = "Long"

print(label)
```

This faulty version prints Short. The second if starts a separate decision and overwrites Invalid. Its else belongs to that second if. Restoring elif makes the second test depend on the first test failing. Moving print inside one branch can also hide output on other paths; indentation affects behavior, not just appearance.

The script rejects negative integers, but it does not validate arbitrary types or check a measured duration. A supplied text value can raise an error during comparison. Keep the stated integer input contract and report the limitation honestly.

Practise with less help: Trace the tested path

Use workbook W2 for the connected example “Branching and boundary tests”. First fill the input and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan · Do · Check

Plan: What is given? Which rule or procedure applies?

Do: Record each intermediate step.

Check: Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Branching and boundary tests. Proposed explanation: “Every condition in the chain always runs”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Give expected labels for −1,0,10,11 and identify the equality boundary for Short.

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

9. Run the investigation

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



PREDICT → TEST → RECORD → EXPLAIN

Trace six pet cards through the same ordered rules. Change only the rule order and compare outputs.

Trace six pet cards through the same ordered rules. Change only the rule order and compare outputs.

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

Test / input: normal

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: boundary

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: missing

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: fresh transfer

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

What did the hardest test teach you that the normal test did not?

Write or draw your reasoning and evidence.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

10. Build your understanding

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce evidence without copying the worked example.



MASTERY LADDER
1 Name it • 2 Trace it • 3 Apply it • 4 Challenge it • 5 Explain limits

1 • NAME: Define the concept without repeating the textbook.

2 • TRACE: Show the information, state, branch, calculation, or evidence path.

3 • APPLY: Solve a different example independently.

4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.

5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.

My strongest evidence of independent understanding:

Write or draw your reasoning and evidence.

11. Connect your project

CUMULATIVE PROJECT

Transparent Book Recommender

Every week adds one inspectable piece. The system must show how it reached a result.



TRANSPARENT BOOK RECOMMENDER

Branch when a reader has a hard constraint such as language, format, or maximum length.

THIS WEEK’S CHECKPOINT Branch when a reader has a hard constraint such as language, format, or maximum length.

SYSTEM MAP Reader need → allowed inputs → transparent rule or score → ranked candidates → explanation → test record → human decision

What will you add, change, or verify in the project this week?

Write or draw your reasoning and evidence.

What should remain under human control?

Write or draw your reasoning and evidence.

RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits. Never confuse a ranking with a fact about a person.

Evidence for your project

Validate recommender requests before ranking items, retaining explicit labels for missing or invalid cases.

12. Show what you know

INDEPENDENT MASTERY PROOF

Explain • Transfer • Test • Defend

Complete this page without copying the worked example. Your reasoning matters more than decoration.



INDEPENDENT PROOF

Build and trace a three-branch pet-care algorithm.

CHALLENGE Build and trace a three-branch pet-care algorithm.

1. My fresh example, trace, calculation, or design:

Write or draw your reasoning and evidence.

2. My evidence that it works (include a boundary or missing-data case):

Write or draw your reasoning and evidence.

3. One limitation, fairness concern, or situation needing human review:

Write or draw your reasoning and evidence.

SELF-CHECK ☐ I explained the mechanism ☐ I used evidence ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

13. Study a complete case

Week 3 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

If pages<100 choose Short; otherwise if pages<=200 choose Medium; otherwise Long.

The question

Classify 99,100,200,201 pages.

Worked explanation

Short, Medium, Medium, Long. Strict and inclusive boundaries and branch order matter.

Explain it in your own words

Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

Test strict and inclusive page boundaries

Evidence or stage · Value or result

Evidence or stage: 99 pages

Value or result: Less than 100 → Short

Evidence or stage: 100 pages

Value or result: 100 to 200 inclusive → Medium

Evidence or stage: 200 pages

Value or result: 100 to 200 inclusive → Medium

Evidence or stage: 201 pages

Value or result: Greater than 200 → Long

If pages<100 choose Short; otherwise if pages<=200 choose Medium; otherwise Long.

Classify 99,100,200,201 pages.

Short, Medium, Medium, Long. Strict and inclusive boundaries and branch order matter.

14. Try a new case independently

Independent case: Build a new sorter independently

Use an integer tiles count. Label a negative value Invalid, zero Empty, 1 through 6 Small and values above 6 Large. Write an ordered if elif else script and predict -1, 0, 1, 6 and 7. Include one print after the chain.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 3 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

If pages are at most 80, choose short; otherwise choose long. Classify books with 79, 80 and 81 pages and explain the boundary.

My answer, with a trace, calculation or supporting evidence

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed input and expected result

Write or draw your reasoning and evidence.

Connect to your project

Branch when a reader has a hard constraint such as language, format, or maximum length.

The evidence I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice evidence together.

Transfer practice from the original programme

Classify lengths with if pages<80 then Short; else if pages<=160 then Medium; else Long. Trace 79,80,160,161. What happens if the <=160 branch is placed first?

Explain the mechanism, evidence and one limit.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed. A later check uses a changed case, not a copied answer.

15. Your lesson route

Start with this question: A fictional reading-time labeler uses one Python if/elif/else chain.

By the end, I can · How I will show it

By the end, I can: Trace the selected branch in an if elif else chain

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

By the end, I can: Use indentation to identify which statements belong to a branch

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

By the end, I can: Test values at and around a decision boundary

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project evidence. Your teacher may spread this lesson across several classes.

Keep your first prediction. Compare it with what the evidence shows and explain any change.

16. Finish and return

Close your notes. Explain this goal in your own words: Trace the selected branch in an if elif else chain

My deciding step, evidence and one remaining question

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an input or assumption and justify your prediction.

Save your workbook evidence. At the next review, try a different case before opening your old answer.